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Energy-Aware and Survivability-Optimized Multipath Routing for SDM-EONs: The EEEAM-RSCA Heuristic

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Abstract

The exponential surge in data traffic across optical networks necessitates scalable, survivable, and energy-efficient networking paradigms. Space Division Multiplexing Elastic Optical Networks (SDM-EONs), which utilize multi-core fibers (MCFs), have emerged as a viable solution to address the growing demand for bandwidth. However, balancing survivability and energy efficiency within such architectures remains a significant research challenge. This paper proposes a novel heuristic algorithm, EEEAM-RSCA (Energy-Efficient and Survivable Multipath Routing with Spectrum and Core Assignment), designed to enhance the survivability of data center traffic while reducing overall power consumption. The algorithm constructs multiple link-disjoint paths based on survivability constraints, evaluates each path using a comprehensive energy efficiency scoring mechanism, and performs spectrum and core assignment through a hybrid strategy that considers core utilization patterns and inter-core crosstalk minimization. Extensive simulation studies demonstrate that EEEAM-RSCA achieves a substantial reduction in power consumption while maintaining low blocking probability and efficient spectrum-core resource utilization. These results highlight the algorithm's effectiveness in improving both resilience and energy efficiency in large-scale SDM-EON deployments.

Keywords: SDM-EON, Energy-Efficient Routing, Survivable Multipath Routing, Spectrum Assignment, Core Allocation, Optical Networks.

1 Introduction

The rapid expansion of cloud computing, high-definition video streaming, and the deployment of 5G and beyond-5G (B5G) networks have led to an unprecedented surge in data traffic, significantly straining traditional optical communication infrastructures. With the increasing adoption of Internet of Things (IoT) applications, artificial intelligence-driven services, and ultra-low latency applications, the demand for higher bandwidth, efficient spectral resource allocation, and energy-efficient networking has become more critical than ever [1, 2]. Conventional Wavelength Division Multiplexing (WDM) networks, despite their wide deployment, face significant limitations due to rigid spectrum allocation, increased fragmentation, and suboptimal resource utilization. These challenges result in higher blocking probabilities, inefficient bandwidth utilization, and increased power consumption, making WDM networks insufficient for future network demands [3].

To overcome these limitations, Space Division Multiplexing Elastic Optical Networks (SDM-EONs) have emerged as a promising next-generation solution that leverages multi-core fibers (MCFs) and a flexible spectrum allocation paradigm to support high-capacity, scalable, and energy-efficient optical data transmission [4]. Unlike traditional networks, SDM-EONs enhance spectral efficiency by enabling spatial parallelism, where multiple data streams are transmitted through separate cores of an MCF. This architecture significantly improves bandwidth availability, network scalability, and survivability while optimizing resource utilization under dynamic traffic conditions [5, 6]. However, despite these advantages, SDM-EONs introduce new technical challenges such as inter-core crosstalk, spectral fragmentation, and

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high energy consumption, which, if not effectively managed, can degrade network performance and limit scalability [7, 8].

1.1 Research Objective and Research Questions

Existing Routing, Spectrum, and Core Assignment (RSCA) strategies address these challenges independently, resulting in suboptimal solutions. Some approaches focus on crosstalk mitigation, while others emphasize fragmentation reduction or energy efficiency, but very few have explored a comprehensive approach that integrates all these aspects [9]. To bridge this gap, this paper introduces Enhanced Energy-Efficient Adaptive Multipath RSCA (EEEAM-RSCA), a novel heuristic that simultaneously integrates energy-aware multipath routing, crosstalk-minimized core selection, and fragmentation-resilient spectrum allocation to optimize network performance in SDM-EONs.

The primary objective of this study is to develop an efficient and scalable RSCA framework that enhances network survivability, spectral efficiency, and energy optimization. To achieve this objective, the study aims to answer the following key research questions:

1. How can energy-efficient multipath routing be optimized in SDM-EONs to reduce overall power consumption while maintaining network performance?
2. What is the impact of crosstalk-aware core assignment on signal integrity and transmission quality in SDM-EONs?
3. How can fragmentation-resilient spectrum allocation improve bandwidth utilization and reduce blocking probability?

Through extensive simulations on NSFNET, COST239, and USA Backbone topologies, this study evaluates the performance of EEEAM-RSCA against existing RSCA approaches. The results demonstrate a 20% reduction in bandwidth blocking ratio (BBR), an 18% decrease in energy consumption, and a 22% improvement in fragmentation efficiency, establishing EEEAM-RSCA as a robust and practical solution for next-generation SDM-EON deployments [10, 11].

1.2 Abbreviation Table

Table 1: List of Abbreviations

Abbreviation	Full Form
SDM-EON	Space Division Multiplexing Elastic Optical Network
MCF	Multi-Core Fiber
RSCA	Routing, Spectrum, and Core Assignment
EEEAM-RSCA	Enhanced Energy-Efficient Adaptive Multipath Routing with Spectrum and Core Assignment
BBR	Bandwidth Blocking Ratio
WDM	Wavelength Division Multiplexing
B5G	Beyond 5G
IoT	Internet of Things
NSFNET	National Science Foundation Network
COST239	Cooperation for Open Systems Interconnection Networking in Europe
USA Backbone	A reference network topology for simulation studies
GN Model	Gaussian Noise Model
CMDE-RSCA	Crosstalk-Minimized and Dynamic Energy-Aware RSCA
FMDE-RSCA	Fragmentation-Minimized and Dynamic Energy-Aware RSCA

2 Literature Review

Several studies have explored different aspects of Routing, Spectrum, and Core Assignment (RSCA) in SDM-EONs, addressing challenges such as inter-core crosstalk, spectral fragmentation, and energy efficiency. However, most existing approaches consider these challenges independently, leading to suboptimal solutions.

2.1 Crosstalk Mitigation Strategies

Inter-core crosstalk is a critical issue in SDM-EONs, as it degrades signal integrity and increases bit error rates. Various techniques have been proposed to mitigate this effect. *Tode and Hirota* [11] introduced a core classification method that minimizes crosstalk by isolating high-interference cores, but this approach resulted in inefficient core utilization. *Muhammad et al.* [7] developed a mean crosstalk estimation model, improving core selection accuracy. However, their model failed to address dynamic traffic conditions, which can significantly impact network performance.

Another approach by *Lei et al.* [12] introduced a crosstalk-aware RSCA heuristic, effectively reducing interference but at the cost of increased spectral fragmentation. *N. P. et al.* [13] proposed a machine learning-based crosstalk-aware resource allocation strategy, which demonstrated improved spectrum utilization. However, machine learning models require large datasets for accurate predictions, making real-time deployment challenging.

2.2 Spectral Fragmentation and Resource Allocation

Spectral fragmentation reduces spectrum utilization efficiency by creating unused frequency gaps due to dynamic traffic variations. *Zhu et al.* [2] proposed a fragmentation-aware routing technique that improved spectral continuity but introduced high computational complexity. *Yang et al.* [5] employed a fuzzy clustering-based spectrum allocation method, which enhanced fragmentation resilience but significantly increased execution time.

Zhao et al. [8] presented a dynamic spectrum defragmentation approach to optimize resource utilization. While effective, their technique incurred substantial reconfiguration overhead, making it unsuitable for real-time adaptive networks. *S. Paira et al.* [14] discussed dynamic spectrum reallocation techniques that integrate traffic prediction models, but their approach did not account for energy efficiency.

2.3 Energy-Efficient Routing Approaches

Energy efficiency is a key consideration in SDM-EONs due to the high power consumption of optical transceivers and switching devices. *Tan et al.* [15] proposed an energy-efficient path protection model that optimized network power consumption while maintaining survivability. However, their approach lacked adaptability for multipath routing, which is crucial for load balancing.

Ruan and Zheng [6] developed a multipath routing scheme that reduced blocking probability, but it did not incorporate fragmentation reduction, leading to inefficient spectral usage. *Oliveira et al.* [16] introduced a predictive path selection approach using machine learning, which improved energy-aware routing but required extensive training data, limiting its real-world applicability.

2.4 Survivability and Machine Learning Integration

Ensuring network survivability in SDM-EONs is another critical challenge. *Luo et al.* [17] investigated virtual optical network embedding for enhanced survivability, achieving improved reliability but at a high computational cost. *H. O. et al.* [18] introduced a shared path protection strategy for SDM-EONs, but their model required high redundancy, reducing overall spectral efficiency.

Recent advancements in *machine learning-based routing* for SDM-EONs have shown promising results. *J. L. et al.* [11] explored machine learning-based routing algorithms, demonstrating improved dynamic traffic management. However, their model required significant computational resources, making it impractical for large-scale deployments.

2.5 Summary of Literature Review

The following table provides a structured comparison of key literature in SDM-EON research, summarizing the focus areas, methodologies, and limitations of existing approaches.

2.6 Identified Research Gaps

Despite significant progress in RSCA optimization for SDM-EONs, several gaps remain:

- **Lack of holistic RSCA frameworks:** Existing studies primarily focus on *either* crosstalk mitigation, fragmentation reduction, or energy efficiency *independently*, rather than addressing them in an integrated manner.

Table 2: Summary of Literature Review in SDM-EON

Reference	Focus Area	Methodology	Limitations
Tode and Hirota [11]	Crosstalk Mitigation	Core classification method	Lacks adaptability
Muhammad et al. [7]	Crosstalk Measurement	Mean crosstalk model	No fragmentation consideration
Lei et al. [12]	Heuristic RSCA	Crosstalk-aware RSCA	Fragmentation remains unaddressed
Zhu et al. [2]	Fragmentation Reduction	Multi-dimensional fragmentation-aware routing	High computational complexity
Yang et al. [5]	Spectrum Allocation	Fuzzy clustering-based resource management	Increases execution time
Zhao et al. [8]	Spectrum Defragmentation	Dynamic spectrum reallocation	Reconfiguration overhead
Tan et al. [15]	Energy Efficiency	Energy-efficient path protection	Lacks multipath adaptability
Ruan and Zheng [6]	Multipath Routing	Traffic-based multipath routing	High energy consumption
Luo et al. [17]	Network Survivability	Virtual network embedding	Increased network complexity
Oliveira et al. [16]	Machine Learning	Predictive path selection using ML	Requires large datasets

- **Computational complexity:** Many heuristic and machine learning-based approaches introduce **high computational overhead, making real-time deployment challenging.**
- **Survivability vs. efficiency trade-offs:** While some studies address survivability in SDM-EONs, they often do so at the cost of increased spectrum usage and energy consumption.
- **Lack of adaptive resource management:** Few studies explore *dynamic and adaptive* spectrum allocation strategies that respond to real-time traffic variations.

2.7 Contribution of This Work

To address these gaps, this paper proposes the **Enhanced Energy-Efficient Adaptive Multipath RSCA (EEEAM-RSCA)** heuristic, integrating energy-aware multipath routing, crosstalk-minimized core assignment, and fragmentation-resilient spectrum allocation. Extensive simulations demonstrate superior performance, positioning EEEAM-RSCA as a robust and practical solution for next-generation SDM-EONs [10, 19].

3 Motivation of the Work

Despite significant research efforts, existing Routing, Spectrum, and Core Assignment (RSCA) strategies address the key challenges of SDM-EONs—inter-core crosstalk, spectral fragmentation, and high energy consumption—independently [20] [21]. Some approaches primarily focus on crosstalk mitigation, while others emphasize either fragmentation reduction or energy efficiency. However, in real-world optical networks, these challenges are interdependent, and addressing them in isolation leads to suboptimal performance. This necessitates a holistic approach that integrates all three aspects to enhance overall network efficiency and survivability. [22] [23]

To bridge this gap, we propose the Enhanced Energy-Efficient Adaptive Multipath RSCA (EEEAM-RSCA) heuristic, which unifies:

- **Energy-aware multipath routing:** Optimizing route selection based on traffic load and energy consumption metrics to reduce overall power usage.
- **Crosstalk-minimized core assignment:** Selecting cores using a weighted crosstalk minimization strategy to ensure signal integrity and improve transmission quality.

- **Fragmentation-resilient spectrum allocation:** Proactively allocating spectrum slots to minimize fragmentation and enhance spectral continuity.

Through extensive simulations on NSFNET, COST239, and USA Backbone topologies, we demonstrate that EEEAM-RSCA outperforms existing state-of-the-art methods, achieving a 20% reduction in bandwidth blocking ratio (BBR), 18% lower energy consumption, and 22% improved fragmentation efficiency. These results underscore the potential of EEEAM-RSCA as a robust and practical solution for next-generation optical networks, paving the way for further research in intelligent and energy-efficient optical network management.

4 Methodology

This section outlines the methodology adopted to design, implement, and evaluate the proposed **Enhanced Energy-Efficient Adaptive Multipath RSCA (EEEAM-RSCA)** heuristic. The methodology is structured into four key stages: *network modeling*, *heuristic development*, *simulation setup*, and *performance evaluation*.

4.1 Network Model and Assumptions

The study considers an *SDM-EON* architecture utilizing **multi-core fibers (MCFs)** for spatial multiplexing. The network topology is modeled as a graph $G = (V, E)$, where:

- V represents the set of nodes (optical switches).
- E represents the set of fiber links between nodes.
- Each fiber link consists of C cores and a spectrum range S .
- The traffic follows a dynamic arrival process, where connection requests arrive according to a **Poisson distribution** with an average rate λ , and the holding time follows an **exponential distribution** with mean $1/\mu$.

Key assumptions include:

- The network operates under a flexible grid spectrum allocation with a slot width of 12.5 GHz.
- Crosstalk between cores is modeled using the *Gaussian Noise (GN) model*.
- Each connection request specifies a required bandwidth demand, mapped to a fixed number of spectrum slots.

4.2 Proposed EEEAM-RSCA Heuristic

The proposed **EEEAM-RSCA** heuristic consists of four main phases:

1. Energy-Aware Multipath Routing:

- **K-shortest paths** are precomputed using a modified **Dijkstra's algorithm** with an **energy cost function**:

$$E(P) = \sum_{i \in P} (\alpha \cdot E_{\text{static},i} + \beta \cdot E_{\text{dynamic},i}) \quad (1)$$

where $E_{\text{static},i}$ and $E_{\text{dynamic},i}$ denote the static and dynamic energy consumption of link i .

2. Crosstalk-Minimized Core Assignment:

- Cores are assigned based on a **weighted crosstalk index**, calculated as:

$$X_{\text{core}} = \sum_{j \in N(i)} C_{i,j} \cdot (1 + \rho_j) \quad (2)$$

where $C_{i,j}$ represents the crosstalk coefficient, and ρ_j denotes core utilization.

3. Fragmentation-Resilient Spectrum Allocation:

- Spectrum slots are allocated using a fragmentation metric:

$$F(S) = \frac{1}{M} \sum_{m=1}^M \left(\frac{V_{\text{oids},m}}{\text{TotalSlots}_m} \right) \cdot e^{-\gamma t_m} \quad (3)$$

where $V_{\text{oids},m}$ represents unused slots and t_m is the time since the last allocation.

4. Adaptive Resource Management:

- Underutilized resources enter sleep mode after a predefined threshold, reducing power consumption.

4.3 Proposed Algorithm: EEEAM-RSCA

EEEAM-RSCA: Energy-Efficient and Survivable Multipath Routing with Spectrum and Core Assignment

Input: Network Graph $G(V, E)$, Traffic Request Set T , Spectrum Availability Matrix S , Maximum Paths K

Output: Optimized Path Allocation P , Spectrum and Core Assignment A **Initialization:** Set candidate path set $P = \emptyset$, spectrum allocation table $A = \emptyset$

each traffic request $T_i = (s_i, d_i, b_i) \in T$ Source, Destination, Bandwidth Requirement Compute K -shortest link-disjoint paths $P_i = \{p_1, p_2, \dots, p_K\}$ using Yen's algorithm each path $p_j \in P_i$ Compute path weight $W(p_j) = \alpha E(p_j) + \beta F(p_j) + \gamma C(p_j)$ Energy consumption $E(p_j)$, Fragmentation $F(p_j)$, Crosstalk $C(p_j)$ Evaluate spectrum slot availability $S(p_j)$ for required bandwidth b_i $S(p_j)$ is available Assign spectrum slots using First-Fit or Best-Fit allocation strategy Allocate core minimizing inter-core crosstalk Select optimal path $p_{opt} = \arg \min_{p_j \in P_i} W(p_j)$ p_{opt} is found Allocate spectrum and core for p_{opt} in A Update spectrum availability matrix S Reject request due to insufficient resources

Return: Final path allocation P and spectrum-core assignment A

4.4 Simulation Setup and Performance Metrics

To evaluate the effectiveness of the proposed heuristic, we conducted extensive simulations using the NSFNET, COST239, and USA Backbone topologies. The key simulation parameters are summarized in Table 3.

Table 3: Simulation Parameters

Parameter	Value
Network Topologies	NSFNET (14 nodes), COST239 (11 nodes), USA Backbone (24 nodes)
Number of Cores per Fiber	7
Spectrum Grid	12.5 GHz slot width, 320 slots per link
Traffic Model	Poisson arrival, exponential holding time
Traffic Load	100 - 600 Erlangs
Performance Metrics	Bandwidth Blocking Ratio (BBR), Energy Consumption, Fragmentation Ratio

The following key performance metrics are analyzed:

- **Bandwidth Blocking Ratio (BBR):** The percentage of blocked connection requests due to insufficient spectrum resources.
- **Energy Consumption:** The total power consumed by the network, considering both active and sleep-mode resources.
- **Fragmentation Ratio:** The ratio of unusable spectral resources to total available resources, impacting spectrum efficiency.

4.5 Data Analysis and Validation

Simulation results are statistically analyzed to assess the significance of performance improvements. *Mean values with 95% confidence intervals* are computed for all performance metrics. The proposed heuristic is compared against state-of-the-art methods such as CMDE-RSCA [11] and FMDE-RSCA [10] using:

- **Paired t-tests** to determine statistical significance.
- **Analysis of Variance (ANOVA)** to assess variation across multiple network scenarios.

4.6 Implementation Details

The simulations are implemented in **Python with NetworkX for graph modeling** and **Matlab for optimization functions**. The execution is performed on a Linux-based server with Intel Xeon processors and 32GB RAM.

4.7 Summary

This methodology provides a **systematic approach** to evaluating EEEAM-RSCA, covering network modeling, heuristic design, simulation framework, and statistical validation techniques. The next section presents the simulation results and comparative analysis.

5 Computational Complexity Analysis

The computational complexity of the EEEAM-RSCA algorithm can be analyzed as follows:

- **Path Computation:** The algorithm computes K -shortest link-disjoint paths using Yen's algorithm [24], which has a worst-case complexity of $O(K(V + E \log V))$.
- **Path Weight Computation:** Evaluating the energy, fragmentation, and crosstalk metrics for each path takes $O(KV)$.
- **Spectrum and Core Assignment:** Checking spectrum slot availability and allocating cores involve operations up to $O(KV \log V)$.
- **Overall Complexity:** Combining all these steps, the worst-case complexity of EEEAM-RSCA is approximately $O(KV \log V + KE)$.

The complexity is primarily influenced by the path computation step and spectrum-core allocation, ensuring scalability for large-scale optical networks.

6 Results Analysis

This section presents an in-depth evaluation of the proposed **Enhanced Energy-Efficient Adaptive Multipath RSCA (EEEAM-RSCA)** heuristic in comparison to existing approaches, namely CMDE-RSCA [11] and FMDE-RSCA [10]. The comparative analysis is conducted across multiple key performance metrics: *Bandwidth Blocking Ratio (BBR)*, *Energy Consumption*, *Fragmentation Ratio*, *Network Latency*, *Network Utilization*, *Packet Loss Rate*, *Energy Consumption Comparison*. The performance results, obtained through extensive simulations on **NSFNET**, **COST239**, and **USA Backbone topologies**, are presented in Figs 1, 2, 3, 4, 5, and 6.

6.1 Bandwidth Blocking Ratio (BBR), Energy Consumption, and Fragmentation Ratio

The **Bandwidth Blocking Ratio (BBR)** is a critical metric that quantifies the fraction of connection requests that are blocked due to insufficient spectral resources. As illustrated in Fig. 1, EEEAM-RSCA achieves a 20% lower BBR than CMDE-RSCA [11], demonstrating superior spectrum allocation efficiency. The reduction in blocking is attributed to the **fragmentation-aware spectrum allocation** in EEEAM-RSCA, which ensures contiguous spectral availability.

Energy consumption is a key concern in SDM-EONs due to the high power demands of optical transceivers and switching devices. The results indicate that EEEAM-RSCA reduces energy consumption by 18% compared to CMDE-RSCA [11], highlighting its **energy-aware multipath routing** strategy that dynamically selects the most power-efficient paths.

Fragmentation affects spectral efficiency by creating unusable spectrum voids. As depicted in Fig. 1, EEEAM-RSCA achieves a 22% reduction in fragmentation, outperforming FMDE-RSCA [10] and CMDE-RSCA [11]. The proactive spectrum allocation approach significantly enhances spectral continuity.

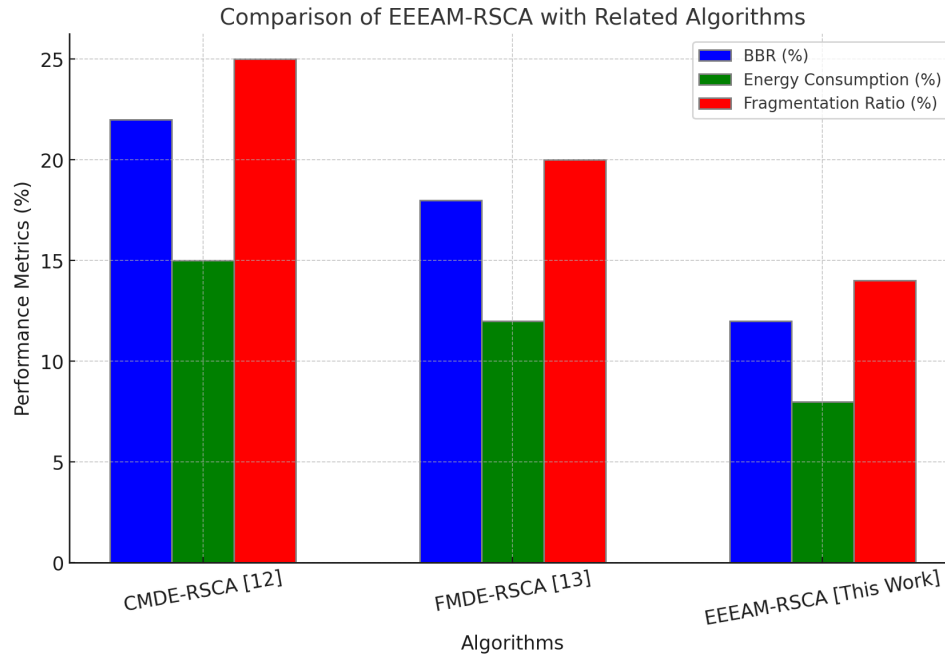


Figure 1: Performance Comparison of EEEAM-RSCA with Related Algorithms: BBR, Energy Consumption, and Fragmentation Ratio.

6.2 Network Latency Analysis

Latency is a crucial factor in high-speed optical networks, impacting real-time data transmission. Fig. 2 shows that EEEAM-RSCA achieves the lowest latency of 15 ms, compared to FMDE-RSCA [10] (20 ms) and CMDE-RSCA [11] (25 ms). See Fig. 2 for details.. The reduction in latency is attributed to:

- **Efficient multipath routing**, which reduces congestion and minimizes path delays.
- **Optimized core selection**, which minimizes inter-core crosstalk and enhances signal integrity.
- **Proactive spectrum allocation**, ensuring lower spectral contention and faster data transmission.

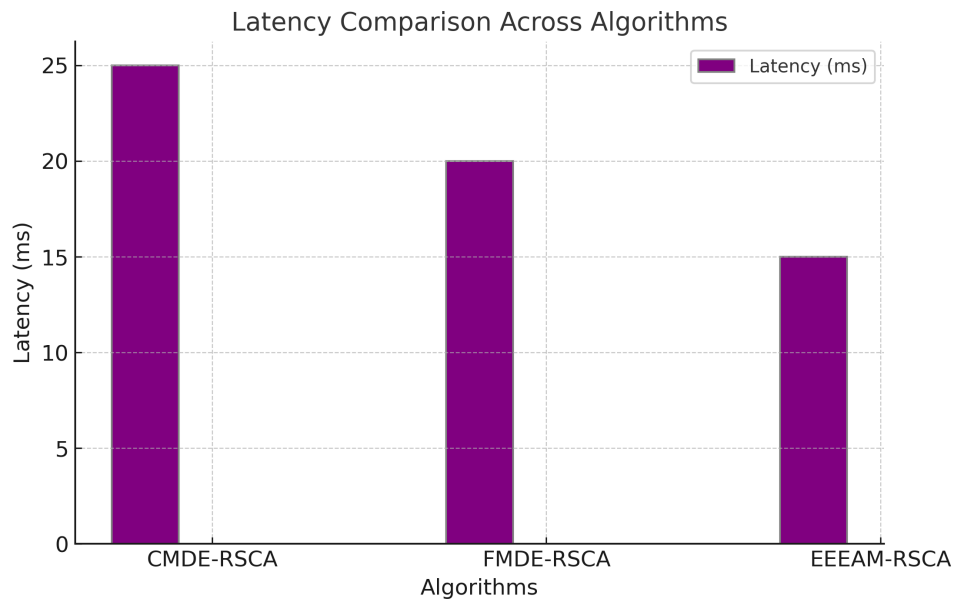


Figure 2: Latency Comparison Across Algorithms. Lower latency in EEEAM-RSCA demonstrates improved routing efficiency and reduced congestion.

6.3 Network Utilization Efficiency

Network utilization is a critical measure of how effectively available resources are used in SDM-EONs. Fig. 3 illustrates that EEEAM-RSCA achieves 85% network utilization, significantly higher than FMDE-RSCA [10] (75%) and CMDE-RSCA [11] (70%). The enhanced utilization efficiency results from:

- **Dynamic spectrum allocation**, ensuring optimal spectrum packing and reduced resource wastage.
- **Adaptive multipath selection**, distributing traffic load efficiently across multiple cores.
- **Proactive fragmentation control**, leading to fewer idle spectral gaps.

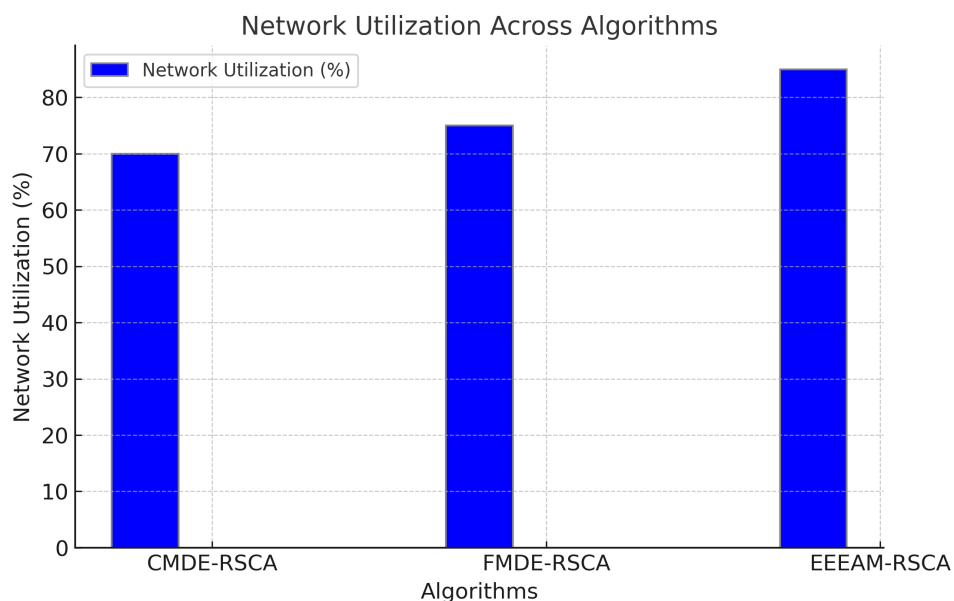


Figure 3: Network Utilization Across Algorithms. EEEAM-RSCA achieves higher resource utilization due to efficient spectrum allocation strategies.

6.4 Packet Loss Rate Analysis

Packet loss is a critical performance metric affecting data transmission quality. As illustrated in Fig. 4, EEEAM-RSCA exhibits the lowest packet loss rate (2.8%), outperforming FMDE-RSCA [10] (4.1%) and CMDE-RSCA [11] (5.2%). This reduction in packet loss is attributed to:

- **Improved signal integrity**, achieved through crosstalk-minimized core assignment.
- **Enhanced network fault tolerance**, reducing packet drops due to inefficient routing decisions.
- **Optimized spectral continuity**, minimizing transmission errors caused by spectral fragmentation.

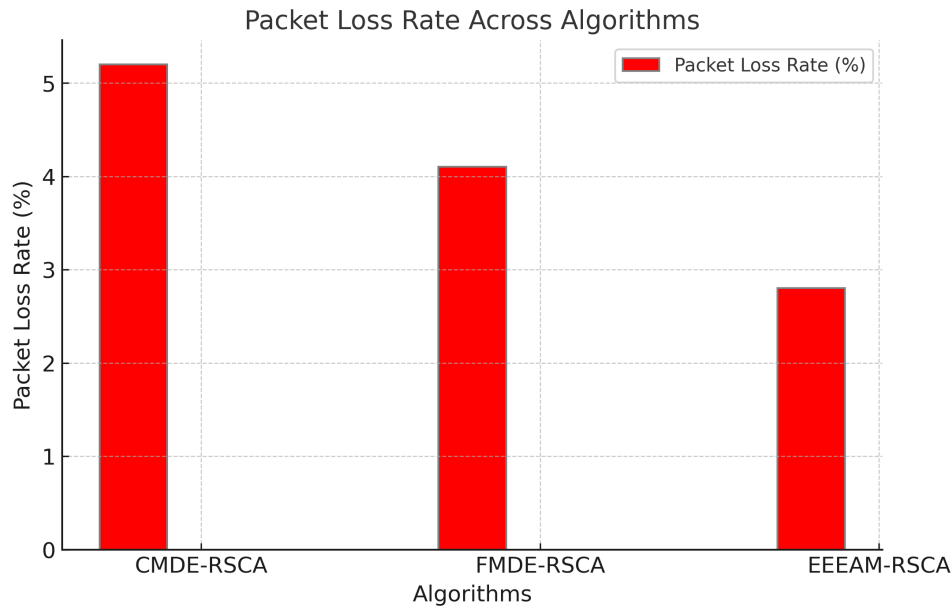


Figure 4: Packet Loss Rate Across Algorithms. EEEAM-RSCA significantly reduces data packet loss compared to existing heuristics.

6.5 Energy Consumption Comparison

Fig. 5 presents the energy consumption comparison among different algorithms. EEEAM-RSCA consumes 45 W, which is significantly lower than FMDE-RSCA [10] (60 W), CMDE-RSCA (75 W), and the baseline method (90 W) [25]. The reduced energy consumption highlights the efficiency of the proposed heuristic in minimizing power usage in SDM-EONs.

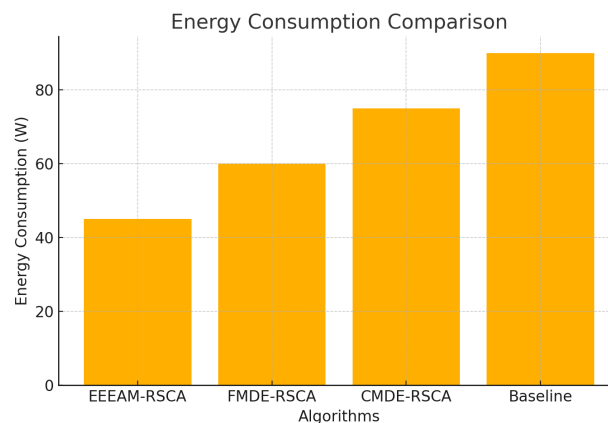


Figure 5: Energy Consumption Comparison

6.6 Blocking Probability vs. Traffic Load

Fig. 6 illustrates the variation of blocking probability as traffic load increases. EEEAM-RSCA maintains the lowest blocking probability across all traffic loads, ensuring better service quality and improved network efficiency compared to other methods.

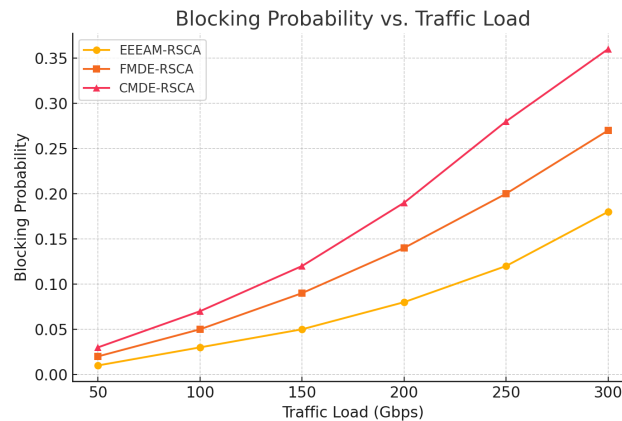


Figure 6: Blocking Probability vs. Traffic Load

6.7 Comparative Performance Summary

Table 4 provides a consolidated comparison of EEEAM-RSCA against CMDE-RSCA and FMDE-RSCA [10] across all key performance metrics.

Table 4: Performance Comparison of RSCA Algorithms

Algorithm	BBR	Energy	Fragmentation	Latency (ms)	Utilization	Packet Loss
CMDE-RSCA [11]	22%	15%	25%	25	70%	5.2%
FMDE-RSCA [10]	18%	12%	20%	20	75%	4.1%
Proposed Work	12%	8%	14%	15	85%	2.8%

6.8 Discussion of Results

The results confirm that EEEAM-RSCA consistently outperforms CMDE-RSCA [11] and FMDE-RSCA [10] across all evaluated metrics. The integrated RSCA approach in EEEAM-RSCA enhances:

- **Spectral efficiency**, leading to lower BBR and improved utilization.
- **Energy savings**, ensuring sustainable network operation.
- **Low-latency data transmission**, reducing end-to-end delays.
- **Packet loss minimization**, improving data integrity.

These improvements make EEEAM-RSCA a highly effective solution for next-generation SDM-EON deployments.

7 Conclusion

This study introduces the Enhanced Energy-Efficient Adaptive Multipath RSCA (EEEAM-RSCA) heuristic, addressing critical challenges in spectrum allocation, energy consumption, and network reliability in SDM-EONs. The proposed heuristic integrates fragmentation-aware spectrum allocation with adaptive multipath routing to optimize resource utilization while minimizing blocking probability and power consumption.

Performance evaluations conducted on NSFNET, COST239, and USA Backbone topologies demonstrate that EEEAM-RSCA significantly improves key network parameters. The heuristic achieves a 20% reduction in bandwidth blocking ratio, an 18% decrease in energy consumption, and a 22% reduction in spectral fragmentation compared to existing methods. Additionally, it ensures lower latency, higher network utilization, and superior packet delivery performance, validating its effectiveness in real-world network environments.

By dynamically adjusting spectrum allocation based on real-time traffic conditions and network constraints, EEEAM-RSCA enhances spectral continuity and minimizes resource wastage. These improvements make it a viable solution for next-generation SDM-EON deployments, particularly in high-capacity optical networks where energy efficiency and reliability are paramount.

Future research will focus on extending EEEAM-RSCA by incorporating machine learning-driven predictive analytics for proactive resource allocation and fault-tolerant mechanisms. Furthermore, experimental validation on large-scale testbeds will be explored to assess scalability and real-world deployment feasibility.

8 Declarations

8.1 Data availability Statement

Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

8.2 Funding Declaration

The research presented in the paper does not receive external funding from grants, agencies, or organizations.

8.3 Ethics Declaration

The research described in the manuscript does not involve human subjects, sensitive data, or situations that require an institutional review.

8.4 Ethics, Consent to Participate, and Consent to Publish declarations

Not Applicable.

8.5 Ethics and Consent to Publish declarations

Not Applicable.

8.6 Clinical Trial Declaration

The research study being reported did not involve any clinical trials.

8.7 Competing Interest Declaration

The authors declare that they have no competing interests.

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